

Uzair Khan

Network Operations Engineer · Satellite Communications

U.S. Citizen · Public Trust · Available immediately

Manassas, VA

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CompTIA CySA+ (CS0-003, through 2027)

SUMMARY

Network Operations Engineer with 8+ years in satellite, enterprise and government network environments, most recently supporting the **DHS/FEMA satellite and IP network at Mount Weather**. **Hands-on iDirect, Hughes and Comtech experience** across GEO and MEO (O3b/SKALA) constellations — terminal restoration, beam switching, P1dB test coordination, site commissioning and real-time capacity management for maritime, aero/IFC and fixed VSAT customers. Layer 2/3 and routing troubleshooting (BGP, OSPF, Cisco, Juniper), plus systems administration with Intune, Azure AD/Entra ID, Active Directory and SCCM. Comfortable working under strict SLAs — carrier coordination, failover, field dispatch, root-cause analysis and clear technical documentation.

iDirect O3b MEO / SKALA DVB-S2 / ModCod P1dB Testing Beam Switching VSAT Commissioning BGP / OSPF
Cisco / Juniper Linux Splunk SolarWinds Intune / Entra ID ServiceNow

PROFESSIONAL EXPERIENCE

NOC Specialist — Satellite Communications

2026

Apex Systems, contractor to AT&T Global Public Sector (DHS/FEMA) · Mount Weather Emergency Operations Center, VA

- **NOC Operations:** Monitored, troubleshoot and configured the DHS/FEMA voice, data and satellite network on a weekend and after-hours rotation, maintaining operational readiness for a mission-critical federal emergency-management network.
- **Satellite Hub Operations:** Operated earth station equipment at an iDirect satellite hub supporting FEMA mobile units; changed space-segment bandwidth assignments for new and changed mission requirements.
- **Troubleshooting & Restoration:** Performed fault isolation, root-cause analysis and service restoration across integrated satellite, LAN/WAN and voice systems; supported network changes and testing with engineering teams.
- **Reporting & Remote Access:** Produced bandwidth usage, traffic analysis and daily schedule reports for the government task order; maintained remote access servers and client systems, including patches and updates.

Systems Administrator / IT Functional Analyst II

2024 – 2025

MDThink – Maryland Department of IT (DoIT): Maryland Benefits (SNAP/DHS) · Linthicum, MD

- **Endpoint Fleet Management:** Managed the lifecycle of 200+ Windows 10/11 devices with Microsoft Intune, SCCM and Autopilot — imaging, role-based access, VPN and application packages for day-one onboarding.
- **Identity & Access:** Administered Azure AD/Entra ID conditional access, group-based licensing and MFA enforcement; owned Jira-based IAM requests, provisioning, de-provisioning and access reviews.
- **Policy & Patch Automation:** Wrote PowerShell scripts deployed via SolarWinds to verify GPO/Intune baseline propagation across hundreds of endpoints; ran firmware and patch deployments under change control.
- **Asset Inventory Automation:** Built a Python/PowerShell scan script to inventory state-owned and vendor devices, feeding a custom Jira schema and automated Confluence reports for the VP and management.
- **Documentation & Dashboards:** Authored KB articles and SOPs adopted organization-wide; built Jira (JQL) dashboards tracking queues, shipping and device lifecycle; prototyped an AI helpdesk chatbot for Google Spaces.
- **Physical Infrastructure:** Audited wall jacks and switch-port assignments, labeled patch panels and maintained network closets; provided on-site AV and network support for executive presentations.

Network Engineer — Network Operations Center (SATCOM)

2022 – 2024

SES Networks · Manassas, VA

24/7 NOC for a global satellite and IP network serving maritime vessels, fixed VSAT and aero (inflight connectivity) across GEO and MEO constellations. First line of response for SATCOM and backbone router issues; primary contact for SKALA (O3b MEO) and VNO customers.

- **Terminal Restoration & Configuration:** Logged into iDirect, Hughes and Comtech terminals over in-band and out-of-band paths (management VLAN, cellular backup, Cisco ASA VPN) to clear faults, push configuration changes and restore customer traffic during service-affecting events.
- **iDirect Platform Operations:** Administered terminals via iVantage NMS and iQ Desktop — Eb/No, Ec/No, RX/TX power, ACM state, symbol rate, BER and burst timing; analyzed modem logs (network_entry, burst_timing, demod_lock) to isolate RF vs. network-layer faults.
- **Beam Switching & Capacity:** Executed manual beam switches for maritime vessels via GUI profiles and modem CLI; rebalanced vessel distribution to relieve beam congestion and managed bandwidth package changes with capacity planning.
- **SKALA / O3b MEO:** Managed SKALA bandwidth packages via GUI and CLI, monitored MEO satellite handoffs and intervened on failed handoff events for SKALA VNO customers.
- **DVB-S2 / RF Management:** Made real-time ModCod adjustments on DVB-S2 forward and return carriers; managed SCPC/MCPC carriers; distinguished rain fade from RFI and escalated spectrum analysis to RF engineering; troubleshoot BUC and remote-site hardware faults, walking site crews through physical checks.
- **Aero Operations:** Monitored live inflight routes through airline ops dashboards, managing bandwidth allocation, beam handoffs and ModCod transitions to hold SLA throughput for cabin IFC.

- **P1dB Test Coordination:** Bridged P1dB transmit power tests between site technicians, ground station RF engineers and the NOC — stepped power increments, documented compression points, locked approved TX ceilings into NMS profiles and cleared carriers for service.
- **Site Commissioning:** Commissioned fixed and maritime VSAT terminals end-to-end — azimuth/elevation verification, modem registration, link-margin validation, P1dB test, activation and as-builts; recommissioned after equipment swaps and dry-dock returns; decommissioned legacy sites.
- **Network Layer & Monitoring:** Monitored OSPF backbone links; ran BGP diagnostics on Juniper MX5 vessel CPE (neighbor state, prefix advertisements, session resets); correlated SNMP, NetFlow and syslog data in SATNMS, Zenoss and Wireshark.
- **Splunk SIEM:** Onboarded 30+ ground station and network log sources; built SPL searches and threshold alerts so recurring fault patterns surfaced faster.
- **Incident & Change Management:** Led incident bridges across satellite ops, terrestrial engineering, ground stations and vendor NOCs; authored RCA reports and customer change-request notices in ServiceNow; mentored junior analysts and wrote commissioning runbooks and P1dB procedures.

Impact: Kept SLA-bound maritime and aero links in service through fast restoration and failover. Splunk onboarding improved fault correlation across 30+ nodes. P1dB bridge coordination helped speed SKALA and VNO customer activations.

Enterprise Network Technician — Network Operations Center

2021 – 2022

ViaPath · Falls Church, VA

- **WAN/LAN Monitoring:** Monitored 100+ correctional facility sites in SolarWinds NPM/NTA; responded to SNMP alerts within 15-minute SLA windows and ran root-cause diagnostics on outages and degradation.
- **SLA & Carrier Coordination:** Worked strict SLAs with on-site contacts and LECs — validated circuit disruptions and communicated ETTR for both carrier-side and internal issues; tracked planned maintenance through CR processes.
- **Failover & Dispatch:** Failed sites over to redundant secondary routes, including ViaSat satellite circuits for remote facilities; opened work orders and dispatched field technicians when remote restoration failed.
- **Site Turn-Up:** Supported new site commissioning and legacy decommissions; gathered and supplied device MAC/IP details to engineering when new devices could not reach the internet or intranet.

Enterprise Account & Operations Support

2017 – 2020

Carfax · Centreville, VA

- Tier 2 support for CARFAX RESTful API and POS integrations; diagnosed data transmission, field mapping and OAuth issues between dealer management systems and vehicle history APIs.
- Remotely deployed integration software on client POS systems; managed tickets in Salesforce and escalated API defects to engineering with reproduction steps and log artifacts.

Retail Operations Manager

2012 – 2016

AT&T Mobility · Oakton, VA

- Used AT&T backend CLI provisioning to activate service, push SIM configurations and resolve carrier-level provisioning failures; activated DIRECTV satellite installs through DSAT and coordinated technician dispatch.
- Configured MDM, enterprise email, VPN and MAM policies for business accounts; led a team of 10 and owned store revenue targets.

TECHNICAL SKILLS

Satellite / RF	iDirect (iVantage, iQ Desktop), Hughes, Comtech CDM, O3b MEO/SKALA, DVB-S2, SCPC/MCPC, ModCod/ACM, P1dB testing, beam switching, VSAT commissioning, HTS/FSS, aero IFC, earth station / hub operations
Networking	BGP, OSPF, EIGRP, MPLS, VLANs, STP, LACP, TCP/IP, DNS, DHCP, NTP, VPN, LAN/WAN/MAN · Cisco ASA/IOS/NX-OS, Juniper MX/EX (JunOS)
Monitoring	SolarWinds NPM/NTA, Splunk, SATNMS, Zenoss, SNMP, NetFlow, syslog, Wireshark, tcpdump
Systems	Windows 10/11, Linux (Ubuntu/WSL, Pi OS), Docker, Intune, SCCM, Autopilot, Entra ID, Active Directory, GPO, VMware user support
Automation / ITSM	Python, PowerShell, bash, Splunk SPL · ServiceNow, Jira (JQL, dashboards), Confluence
Documentation	SOP/KB authoring, RCA reports, change requests, commissioning runbooks, as-builts

PROJECTS

Home Lab & AI-Assisted Operations (ongoing): Raspberry Pi + WSL/Ubuntu lab running Docker services (Pi-hole/Unbound DNS, Tailscale, Nextcloud, Gitea, monitoring); use Claude and AI tools to automate ops, scripting and docs; worked a Pi memory-pressure outage and planned a migration.

TankPit Discord Bot & RAG AI System · github.com/uk0430: Hybrid RAG + SQL AI engine on local LLM inference (Mistral 7B / llama.cpp) with async scrapers feeding a 13,250-player SQLite and vector store. Python, asyncio, Docker.

EDUCATION, CERTIFICATIONS & CLEARANCE

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Strayer University — B.S. Information Technology, expected 2028 (enrolled; completing prerequisites at NVCC) · **NVCC** — A.A.S. Business Administration